

Suppose x is a single case then the solution for Euler-Lagrange equation is a 2nd order differential equation. We have two integration constants and these integration constants can be found from initial and final condition of problem. Suppose x is n variable case then we shall get $2n$ constants and those can be got from initial and final conditions.

In conclusion Euler-Lagrange equation is necessary condition for optimisation of a functional.

Remarks:

- 1) Euler-Lagrange equation is non-linear differential equation since product of $x''(t)$, $x'(t)$ and $x(t)$ may appear in equation.
- 2) Solution involve Two-point boundary value problem.
- 3) Solution through numerical technique and solution through analytic methods are very difficult.
- 4) If Euler-Lagrange equation is not satisfied for any function, this indicates the optimum does not exist for that functional.

Example (courtesy Wikipedia)

Find the real-valued function $y(x)$ on the interval $[a,b]$, such that $y(a) = c$ and $y(b) = d$, for which the path length along the curve traced by y is as short as possible.

$$s = \int_a^b \sqrt{dx^2 + dy^2} = \int_a^b \sqrt{1 + y'^2} dx,$$

the integrand function being $L(x, y, y') = \sqrt{1 + y'^2}$

The partial derivatives of L are:

$$\frac{\partial L(x, y, y')}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}} \text{ and } \frac{\partial L(x, y, y')}{\partial y} = 0.$$

By substituting these into the Euler-Lagrange equation, we obtain

$$\begin{aligned} \frac{d}{dx} \frac{y'(x)}{\sqrt{1 + (y'(x))^2}} &= 0 \\ \frac{y'(x)}{\sqrt{1 + (y'(x))^2}} &= C = \text{constant} \\ \Rightarrow y'(x) &= \frac{C}{\sqrt{1 - C^2}} =: A \\ \Rightarrow y(x) &= Ax + B \end{aligned}$$

In other words, the function must have a constant first order derivative, and hence, its graph is a straight line.

Check Your Progress 1

- 1) Let J be the functional defined by

$$J(y) = \int_0^1 (y'^2 + y^2 + 4ye^t) dt$$

With boundary conditions $y(0) = 0$ and $y(1) = 1$ Find the extremal(s) in interval $[0, 1]$ for $J(y)$

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22.4 EULER EQUATION IN SPECIAL CASES

Here, we discuss two special cases of EL equation (I) with no Explicit $x(t)$ Dependence and (II) with no Explicit 't' Dependence.

22.4.1 Case I: No Explicit $x(t)$ Dependence

Suppose that the functional is of the form

$$J(x(t)) = \int_{t_0}^{t_f} v(x'(t), t) dt$$

Where the variable $x(t)$ does not appear explicitly in integrand. Evidently the Euler-Lagrange equation reduces to

$$\frac{\partial v(\cdot)}{\partial x'(t)} = C_1 \quad \dots(22.13)$$

Where C_1 is a constant of integration. Now $\frac{\partial v(\cdot)}{\partial x'(t)}$ is a known function of t and $x'(t)$ so that the above equation is first order differential equation in form of $x(t)$. In principle this equation is solvable for $x'(t)$ provided $\frac{\partial^2 v(\cdot)}{\partial x'^2(t)} \neq 0$ (one can invoke a variant of the implicit function theorem), so that this equation could be recast in form

$$x'(t) = g(t, C_1) \quad \dots(22.14)$$

for some function g and then integrand.

22.4.2 Case II: No Explicit 't' Dependence

Another simplification is available when the integrand does not contain the independent variable 't' explicitly.

Theorem: Let J be functional of the form

$$J(x(t)) = \int_{t_0}^{t_f} v(x(t), x'(t)) dt \text{ and define the function H by}$$

$$H(x(t), x'(t)) = x'(t) \left[\frac{\partial v(\cdot)}{\partial x'(t)} - V(\cdot) \right] \quad \dots(22.15)$$

Then, H is constant along any extremal $x^*(t)$.

Proof: Suppose $x^*(t)$ an extremal for $J(\cdot)$ Now,

$$\begin{aligned} \frac{d}{dt} H(x^*(t), x'^*(t)) &= \frac{d}{dt} \left(x'^* \frac{\partial V(\cdot)}{\partial x'^*} - V(\cdot) \right) \\ &= x''^* \frac{\partial V(\cdot)}{\partial x'^*(t)} + x'^* \frac{d}{dt} \frac{\partial V(\cdot)}{\partial x'^*(t)} - \left(x'^* \frac{\partial V(\cdot)}{\partial x(t)} + x''^*(t) \frac{\partial V(\cdot)}{\partial x'^*(t)} \right) \\ &= x'^* \left(\frac{d}{dt} \frac{\partial V(\cdot)}{\partial x'^*(t)} - \frac{\partial V(\cdot)}{\partial x(t)} \right) \quad \dots\dots \text{same as equation (22.15)} \end{aligned}$$

And since x^* is an extremal, the Euler-Lagrange equation is satisfied. Hence,

$$\frac{d}{dt} H(x^*(t), x'^*(t)) = 0 \quad \dots\dots(22.16)$$

Consequently, H must be a constant along an extremal.

Note that the function H depends on $x(t)$ and $x'(t)$ and thus the equation

$H(x(t), x'(t)) = \text{constant}$ is a first order differential equation for $x^*(t)$

22.5 SECOND ORDER CONDITION FOR OPTIMALITY

This is called also sufficient condition and this condition determines whether the optimal is minimal or maximal. The sufficient condition is same as the condition for multivariable static optimisation.

Earlier we have set $\nabla J = \delta J + \delta J^2$

We have to optimise the functional we have seen 2nd variation of functional in Taylor series expansion of incremental of functional i.e. δJ^2 (2nd derivative of moment)

$$\begin{aligned} \delta J^2 &= \frac{1}{2} \left\{ \int_{t_0}^{t_f} \left[\partial x(t) \cdot \frac{\partial^2 V(\cdot)}{\partial x^2(t)} \partial x(t) + 2 \partial x(t) \frac{\partial^2 V(\cdot)}{\partial x(t) \partial x'(t)} \partial x'(t) \right. \right. \\ &\quad \left. \left. + \partial x'(t) \frac{\partial^2 V(\cdot)}{\partial x'^2(t)} \partial x'(t) \right] dt \right\} \\ &= \frac{1}{2} \left\{ \int_{t_0}^{t_f} [\partial x(t) \quad \partial x'(t)] \begin{bmatrix} \frac{\partial^2 V(\cdot)}{\partial x^2(t)} & \frac{\partial^2 V(\cdot)}{\partial x(t) \partial x'(t)} \\ \frac{\partial^2 V(\cdot)}{\partial x(t) \partial x'(t)} & \frac{\partial^2 V(\cdot)}{\partial x'^2(t)} \end{bmatrix} \begin{bmatrix} \partial x(t) \\ \partial x'(t) \end{bmatrix} dt \right\} \quad \dots\dots(22.17) \end{aligned}$$

Let the above matrix is A

$\partial^2 J > 0$ if A is positive definite at x^* then $J^*(x^*(t))$ is minimum. So what we have to do is use 1st order condition to evaluate x^* then calculate value of

matrix 'A' at x^* . If the matrix is positive definite then x^* is path that minimizes $J(\cdot)$ by similar analogy

When $\partial^2 J < 0$ if A is negative definite matrix at x^* .

22.6 LEGENDRE CONDITION

Consider a continuously differentiable function $x(t)$ that maximizes $\int_{t_0}^{t_f} V(x, x', t) dt$ subject to $x(t_0) = x_0$, $x(t_f) = x_f$,

Where V is twice continuously differentiable in its three arguments. Then the Legendre condition states that for a maximizing path x^* must satisfy not only the Euler-Lagrange equation but also the Legendre condition i.e.,

$$\frac{\partial^2 V(\cdot)}{\partial x'^2} \leq 0 \quad \text{.....(22.18)}$$

$$\text{For a minimizing path } \frac{\partial^2 V(\cdot)}{\partial x'^2} \geq 0 \quad \text{.....(22.19)}$$

However, if the integrand $V(\cdot)$ is concave in (x, x') a path that satisfies the Euler equation is maximizing. If the integrand is convex in (x, x') , then a path satisfying the Euler equation is minimizing. Thus, the Euler equation together with concavity (convexity) of the integrand in (x, x') are sufficient for a maximum (minimum).

22.7 NECESSARY AND SUFFICIENT CONDITION FOR OPTIMALITY

Let's consider a functional $J(\cdot)$ which is defined as $\int_{t_0}^{t_f} V(x(t), x'(t), t) dt$ we wish to find $x^*(t)$ that in turn optimise (minimize or maximize) the objective functional $J(\cdot)$, where t_0 = initial time. t_f = final time

Further $x(t_0) = x_0$ and $x(t_f) = x_f$

We assume $V(\cdot)$ is continuous and twice differentiable in all its arguments.

For minimization,

$$\text{Necessary condition } \frac{\partial V(\cdot)}{\partial x(t)} - \frac{d}{dt} \left(\frac{\partial V(\cdot)}{\partial x'(t)} \right) = 0 \quad \text{.....(22.20)}$$

This is Euler-Lagrange equation. Solve the above equation to get $x^*(t)$

$$\text{sufficient condition: The matrix } A = \begin{bmatrix} \frac{\partial^2 V(\cdot)}{\partial x^2(t)} & \frac{\partial^2 V(\cdot)}{\partial x(t) \partial x'(t)} \\ \frac{\partial^2 V(\cdot)}{\partial x(t) \partial x'(t)} & \frac{\partial^2 V(\cdot)}{\partial x'^2(t)} \end{bmatrix} \quad \text{.....(22.21)}$$

The matrix A evaluated at x^* is positive definite.

For maximization,

Necessary condition: Same as necessary condition for minimization

Sufficient condition: The matrix A evaluated at x^* is negative definite.

A matrix M is positive definite if $x^T M x > 0$ where x is $n \times 1$ nonzero column vector

A matrix M is negative definite if $x^T M x < 0$

Where x is $n \times 1$ non zero column vector

22.8 ISOPERIMETRIC PROBLEM

There are types of problems for which we seek optimal value of functions which are subjected to integral constraint, in addition to ‘ordinary’ boundary conditions. Such problems are called **isoperimetric problems**. Let $G(x, y, x') dx = L$. The condition $J(y) = L$ is called the integral constraint for some constant λ .

An optimisation problem may be subject to an integral constraint as follows:

Consider $J() = \int_{t_0}^{t_f} V(x, x', t) dt$

subject to $\int_{t_0}^{t_f} F(x, x', t) dt = B, x(t_0) = x_0, x(t_f) = x_f$

where v and F are twice differentiable continuous function and B is a given number. So solve this category of problem we append constraint to $J(.)$ by an undermined Lagrange multiplier λ . Hence, we consider our objective function as

$$\begin{aligned} J(.) &= \int_{t_0}^{t_f} [V(x, x', t) - \lambda F(x, x', t)] dt + \lambda B \\ &= \int_{t_0}^{t_f} S(.) dt + \lambda B \end{aligned} \quad \dots\dots(22.22)$$

The necessary condition for optimality is

$$\frac{\partial S(.)}{\partial x} - \frac{d}{dt} \left[\frac{\partial S(.)}{\partial x'} \right] = 0 \quad \dots\dots(22.23)$$

We solve this above equation to get $x^*(\lambda, c_1, c_2)$ i.e. we get an equation with three unknown: λ, c_1, c_2 use this function of x^* to solve $\int_{t_0}^{t_f} F(x, x', t) dt = B$ use boundary value condition $x(t_0) = x_0$ and $x(t_f) = x_f$ to get C_1, C_2 and λ .

Check Your Progress 2

1) Find extremals for

$$\int_0^1 \{[x'(t)]^2 + 10x(t).t\} dt$$

subject to $x(t_0) = 2, x(t_f) = 3$

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2) Find the extremal for the functional

$$J(.) = \int_{t_0}^{t_f} (2x_t'^2 + 24x(t).t) dt,$$

Boundary condition $x(t_0)=0$, $-x(t_f)=2$

$$t_0 = 0 \quad t_f = 2$$

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22.9 FREE END VALUE PROBLEMS AND TRANSVERSALITY CONDITION

Until now we have been dealing with the specific optimisation problems where both ends are fixed. Now we consider a general case where one end point is fixed while other is free.

Again our functional is $J(.) = \int_{t_0}^{t_f} V(x(t), x'(t), t) dt$

We need to find trajectory of $x(t)$ so that $J()$ is optimised.

Refer to the Fig. 22.1.

Now as per our problem A is fixed and B is free. In other word t_f is free.

With little mathematical calculation as we did in theorem now the 1st variation of functional is

$$\delta J = \int_{t_0}^{t_f} \left[\frac{\partial V(\cdot)}{\partial x(t)} - \frac{d}{dt} \left(\frac{\partial V(\cdot)}{\partial x'(t)} \right) \right]_* \delta x(t). dt + \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* \cdot \delta x(t) \Big|_{t=t_f} + V(t, x^*(t), x'^*(t)) \Big|_{t=t_f} \cdot \delta t_f \quad \text{.....(22.24)}$$

For functional to be optimal

We have to use Euler-Lagrange equation

$$\frac{\partial V(\cdot)}{\partial x(t)} - \frac{d}{dt} \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right] = 0 \quad \text{.....(22.25)}$$

$$\text{So } \delta J \approx \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* \delta x(t) + V(x(t), x'(t), t) \Big|_{*, t=t_f} \cdot \delta t_f$$

So, we have four cases

Case I

When t_f and x_f are free we need to solve 3 equations

$$\frac{\partial V(\cdot)}{\partial x(t)} - \frac{d}{dt} \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right] = 0$$

$$\text{and } \left[V(\cdot) - \frac{\partial V(\cdot)}{\partial x'(t)} \cdot x'(t) \right] \Big|_{t=t_f} = 0 \quad \text{.....(22.26)}$$

$$\text{and } \frac{\partial V(\cdot)}{\partial x'(t)} \Big|_{t=t_f} = 0 \quad \text{.....(22.27)}$$

Case II

When t_f is fixed x_{t_f} is free

hence $\delta t_f = 0$

here we need to solve equation (22.25) and (22.27) to get solution. No need of equation (22.26)

Case III

When t_f is free and x_{t_f} is fixed.

Hence $\delta x_f = 0$

Then we need to solve (22.5) and (22.6) to obtain optimal path. No need of equation (22.27).

Case IV

When end point B is restricted to a curve $g(t)$

Then we need to solve equation (22.25) along

$$\left[V(.) - \frac{\partial V(.)}{\partial x'(t)} (\dot{x}'(t) - g'(t)) \right]_{t=t_f} = 0 \quad \dots(22.28)$$

Equation (22.26), (22.27), (22.28) are called transversality equations for corresponding model. In other words, transversality equations are additional end point restriction imposed by the model constraints in order to solve the respective model.

22.10 DISCRETE TIME DYNAMIC OPTIMISATION

We assume that time evolves in a discrete way, meaning that $t \in \{0, 1, 2, \dots\}$ that is $t \in \mathbb{N}_0$

The model is described by at least two categories of variables that evolve along time: a state variable x_t and a control variable u_t . More on state and control variable in next chapter. Although discrete time dynamic optimisation problems can be solved by Lagrange multiplier method (Ref. Chow, 1997), we here adopt dynamic programming method to solve the problem. Consider a two variable discrete optimisation problem

$$J(.) = \sum_{t_1 \& t_2 = t_0}^{t_1 \& t_2 = t_f} f(x_{t_1}, x_{t_2}) \quad \dots(22.29)$$

Where $t_0 = 0$ and $t_f = N$

If x_1 and x_2 are independent in nature, the optimum of $J()$ can be thought of as sum of optimum of $(N-1)$ processes. $J(.)$ which can be thought of as sum $(N-1)$ static optimisation. Therefore, we can safely jump to constrained discrete dynamic optimisation.

22.10.1 Assumptions

- Timing the state variable x_t is usually a stock and is measured at the beginning of period t and the control U_t is usually a flow and is measured at the end of period t .
- Horizon on some text it is referred as planning horizon or time horizon and that can be finite or is infinite ($T = \infty$). The second case more common.
- Objective functional there is an intertemporal utility function which is additively separable, stationary and involves time-discounting (impatience)

$$\sum_{t=0}^T \beta^t f(u_t, x_t) \quad \dots\dots(22.30)$$

- $0 < \beta < 1$, model impatience as $\beta^0 = 1$ and $\lim_{t \rightarrow \infty} \beta^t = 0$
- $f(\cdot)$ is well behaved: it is continuous, continuously differentiable and concave in (U, x) .
- The economy described by an autonomous difference equation $x_{t+1} = g(x_t, U_t)$

Where $g(\cdot)$ is autonomous continuous, differentiable concave. Then, the difference equation verifies the condition for existence and uniqueness of solutions.

- The non-Ponzi game condition holds:
 $\lim_{t \rightarrow \infty} \beta^t x_t \geq 0$ holds for a discount factor $0 < \beta < 1$
- There may be some side condition such as $x_t \geq 0$ $U_t \geq 0$, which may produce corner solutions. We deal only with the case in which the solutions are interior (or the domain of the variables is open).

These assumptions are formalized as optimal control problems;

Definition 1: The simplest optimal control problem

Find $\{U_t^* x_t\}_{t=0}^T$ which solves

$$\max_{\{U_t\}_{t=0}^T} \sum_{t=0}^T \beta^t f(U_t, x_t) \quad \dots\dots(22.31)$$

$x_{t+1} = g(x_t, U_t)$ for x_0, x_T given and T is free

Definition 2: The free terminal state optimal control problem: Find $\{U_t^* x_t\}_{t=0}^{T-1}$ which solves

$$\max_{\{U_t\}_{t=0}^T} \sum_{t=0}^T \beta^t f(U_t, x_t) \text{ such that} \quad \dots\dots(22.32)$$

$U_t \in U$ and $x_{t+1} = g(x_t, U_t)$

For x_0, T is given and x_T is free. If $T = \infty$, we have infinite horizon discounted optimal control problem

Feasible candidate solutions: path of $\{x_t, U_t\}$ that verify $x_{t+1} = g(x_t, U_t)$ x_0 is given for any choice of $\{U_t \in U\}_{t=0}^T$

Necessary and sufficient condition: Hamilton – Jacobi- Bellman (HJB) equation gives a necessary and sufficient condition for optimality of a control with respect to objective functional. In general if the behavioral functions $f()$ and $g()$ are well behaved (continuous, continuously differentiable and concave) then necessary conditions are also sufficient.

Next, we discuss another approach to dynamic optimization, viz., Dynamic Programming.

22.10.2 Dynamic Programming

The principle of dynamic programming (Richard Bellman 1957) state that an optimal trajectory has the following property: for any given initial values of the x_t and for a given value of x_t and U_t in the beginning any period, the U_t should be chosen optimally for the remaining period, if we take the optimal values of the x_t which result from the previous optimal decisions.

We next follow a heuristic approach for deriving necessary condition for problem following the principle of dynamic programming.

Finite horizon: Assume that we know a solution optimal control $\{U^* x_t^*\}_{t=0}^T$

Definition: Value function for time τ

$$V_{T-\tau}(x_\tau) = \sum_{t=\tau}^T \beta^{t-\tau} \int (U_t^*, x_t^*) \quad \dots\dots(22.33)$$

Proposition 1: Given an optimal solution to the optimal control problem solution optimal control $\{U^* x_t^*\}_{t=0}^T$ then it verifies Hamilton- Jacobi Equation.

$$V_{T-\tau}(x_\tau) = \max_{U_{(t)}} \{f(x_t U_t) + \beta V_{T-t-1}(x_{t+1})\}$$

In literature, this is called HJB equation, so, simply Bellman equation

$$\begin{aligned} V_T(x_0) &= \sum_{t=0}^T \beta^t \int (U_t^*, x_t^*) \\ &= \max_{\{U_{(t)}\}_{t=0}^T} \sum_{t=0}^T \beta^t \int (U_t^*, x_t^*) \\ &= \max_{\{U_{(t)}\}_{t=0}^T} (f(x_0 U_0) + \beta f(x_1 U_1) + \beta^2 f(x_2 U_2) \dots) \\ &= \max_{\{U_{(t)}\}_{t=0}^T} \left(f(x_0 U_0) + \beta \sum_{t=1}^T \beta^{t-1} f(x_t U_t) \right) \\ &= \max_{U_0} \left(f(x_0 U_0) + \beta \max_{\{U_{(t)}\}_{t=1}^T} \sum_{t=1}^T \beta^{t-1} f(x_t U_t) \right) \end{aligned}$$

By principle of dynamic programming then

$$V_T(x_0) = \max_{U_0} \{f(x_0 U_0) + \beta V_{T-1}(x_1)\}$$

We can apply the same idea for the value function for any time $0 \leq t \leq T$ to get equation $V_{T-t}(x_t) = \max_{u_t} \{f(x_t, u_t) + \beta V_{T-t-1}(x_{t+1})\}$ which holds for feasible solutions, i.e., verifying

$$x_{t+1} = g(x_t, u_t) \text{ and given } x_0. \quad \text{.....(22.34)}$$

Intuition: we transform the maximization of a functional into a recursive two period problem. We solve the control problem by solving the HJB equation. To do this we need to find the sequence $\{V_T, \dots, V_0\}$ through the recursion

$$V_{T+1}(x) = \max_U \{f(x, U) + \beta V_T(g(x, U))\} \quad \text{.....(22.35)}$$

Infinite Horizon: For infinite horizon discounted optimal control problem the limit function $V = \lim_{j \rightarrow \infty} V_j$ is independent of j so HJB equation becomes

$$V(x) = \max_U \{f(x, U) + \beta V(g(x, U))\} = \max_U H(x, U) \quad \text{.....(22.36)}$$

Properties of the value function. It is usually difficult to get the properties of $V()$. In general continuity is assured but not differentiability (this is a subject for advanced course on DP), see stokes and Lucas (1989).

If some regularity conditions hold, we may determine the optimal control through the optimality condition

$$\frac{\partial H(x, U)}{\partial U} = 0 \quad \text{.....(22.37)}$$

If $H(.)$ is C^2 (twice continuously differentiable) then we get the policy function

$$U^* = h(x)$$

which gives an optimal rule for changing the optimal control, given the state of the economy. If we can determine (or prove that there exists such a relationship) then we say that our problem is recursive. In this case the HJB equation becomes non-linear functional equation

$$V(x) = f(x, h(x)) + \beta V[g(x, h(x))] \quad \text{.....(22.38)}$$

Solution to HJB is by numerical method (most cases) by value function iteration.

22.10.3 The Cake Eating Problem

Example: The cake eating problem (Refer to the section 22.2.2)

We restate the problem as $\max \sum_{t=0}^T \ln(C_t)$

Subject to $W_{t+1} = W_t - C_t$ and $W_0 > 0$

The steps we follow for solving the problems are as follows:

- 1) Set up Bellman equation
- 2) Take a guess for what $v(W_t)$ is and use that in our Bellman equation (use that for $V(x+1)$ in our Bellman equation, just change the time subscript).